# 命名空間和交會點
## ONTAP 9

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# 命名空間和交會點

## 命名空間與交會點總覽

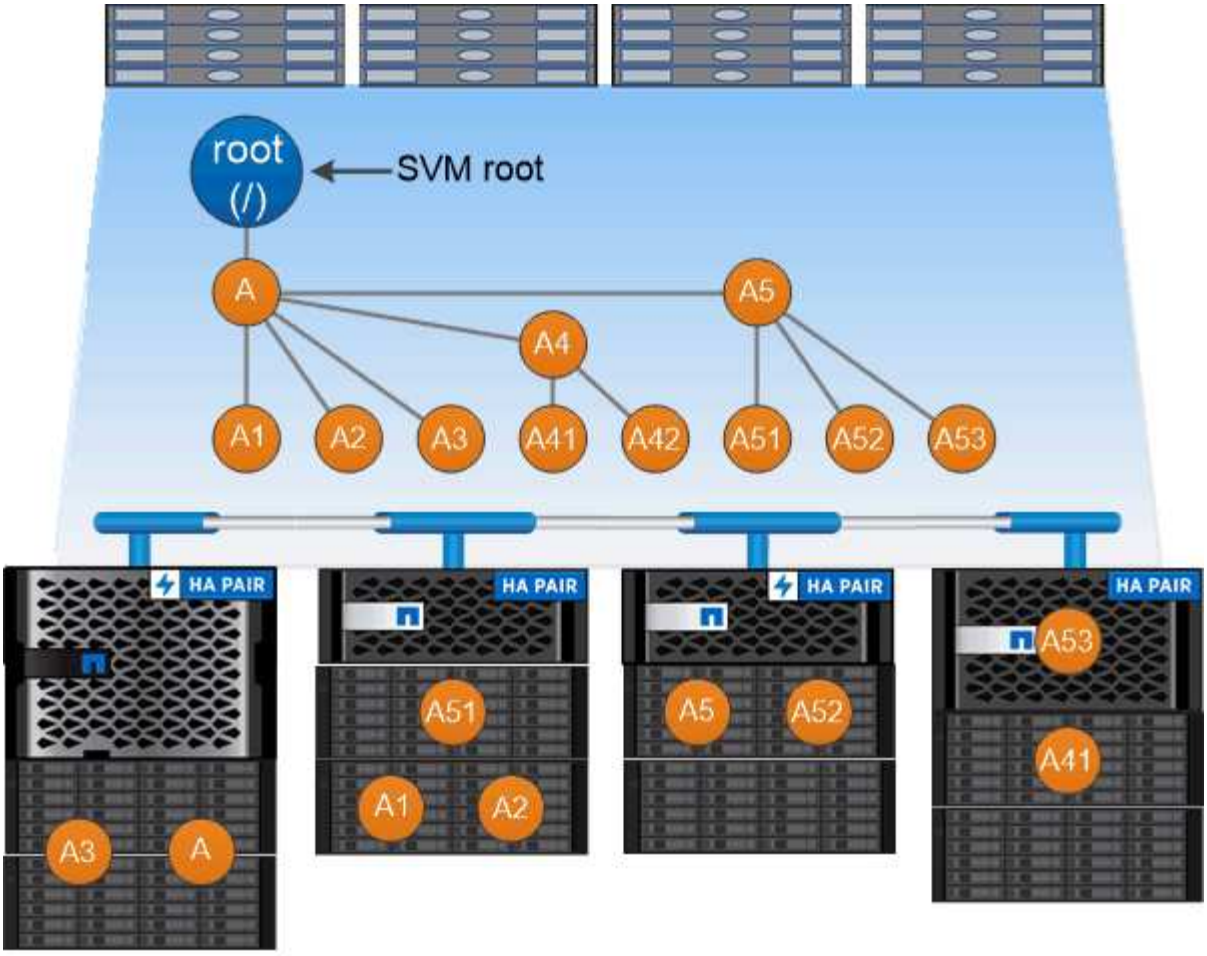
NAS *namespacity*_是一個邏輯群組、集合在_交會點、以建立單一檔案系統階層架構。具有足夠權限的用戶端可存取命名空間中的檔案、而無需指定檔案在儲存設備中的位置。未分段的磁碟區可位於叢集中的任何位置。

NAS用戶端不會掛載每個包含感興趣檔案的磁碟區、而是掛載NFS _EX出口_或存取SMB共享區。_匯出或共用區代表整個命名空間或命名空間內的中繼位置。用戶端只會存取裝載於其存取點下方的磁碟區。

您可以視需要將磁碟區新增至命名空間。您可以直接在父磁碟區交會下方或磁碟區內的目錄上建立交會點。磁碟區的「vol3」連接路徑可能是「/vol1/vol2/vol3」或「/vol1/dir2/vol3」、甚至是「/dir1/dir2/vol3」。路徑稱為_junction路徑。_

每個SVM都有一個獨特的命名空間。SVM根磁碟區是命名空間階層架構的起點。

> 為了確保資料在節點中斷或容錯移轉的情況下仍然可用、您應該為SVM根磁碟區建立_load-sharing mirror _複本。




*A namespace is a logical grouping of volumes joined together at junction points to create a single file system hierarchy.*

下列範例建立名為「home4」的磁碟區、位於SVM VS1、其交會路徑為「/eng/hHome」：

```
cluster1::> volume create -vserver vs1 -volume home4 -aggregate aggr1
-size 1g -junction-path /eng/home
[Job 1642] Job succeeded: Successful
```

# 典型的NAS命名空間架構是什麼

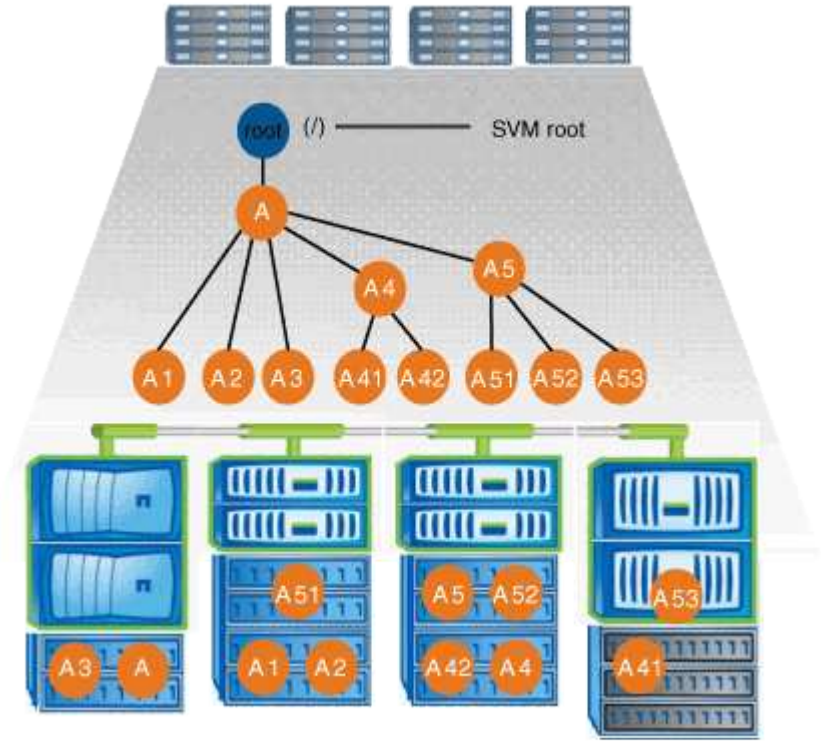
## 您可以在建立SVM名稱空間時、使用幾種典型的NAS命名空間架構。您可以選擇符合業務和工作流程需求的命名空間架構。

命名空間的頂端永遠是根磁碟區、以斜槓（/）表示。根目錄下的命名空間架構可分為三個基本類別：

- 單一分支樹狀結構、只有一個連接點可連接至命名空間的根
- 多個分支樹狀結構、並有多個交會點指向命名空間的根目錄
- 多個獨立磁碟區、每個磁碟區都有一個指向名稱空間根的獨立交會點

## 具有單一分支樹狀結構的命名空間

具有單一分支樹狀結構的架構具有單一插入點、可插入SVM命名空間的根目錄。單一插入點可以是輔助磁碟區、也可以是根目錄下的目錄。所有其他磁碟區都會安裝在單一插入點下方的交會點（可以是磁碟區或目錄）。

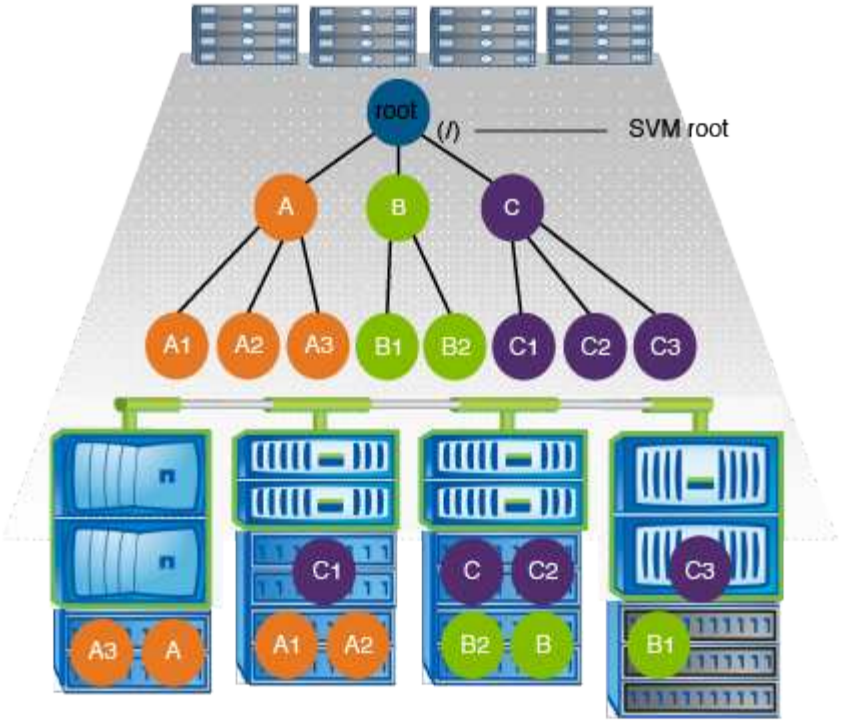



例如、具有上述命名空間架構的典型Volume交會組態可能類似下列組態、其中所有磁碟區都會連結到單一插入點下方、亦即名為「data'」的目錄：

```
                     Junction                          Junction
Vserver Volume       Active   Junction Path            Path Source
------- ------------ -------- --------------------     -----------
vs1     corp1        true     /data/dir1/corp1         RW_volume
vs1     corp2        true     /data/dir1/corp2         RW_volume
vs1     data1        true     /data/data1              RW_volume
vs1     eng1         true     /data/data1/eng1         RW_volume
vs1     eng2         true     /data/data1/eng2         RW_volume
vs1     sales        true     /data/data1/sales        RW_volume
vs1     vol1         true     /data/vol1               RW_volume
vs1     vol2         true     /data/vol2               RW_volume
vs1     vol3         true     /data/vol3               RW_volume
vs1     vs1_root     -        /                        -
```

## 具有多個分支樹狀結構的命名空間

具有多個分支樹狀結構的架構具有多個插入點、可插入到SVM命名空間的根目錄。插入點可以是輔助磁碟區、也可以是根目錄下的目錄。所有其他磁碟區都會安裝在插入點下方的交會點（可以是磁碟區或目錄）。

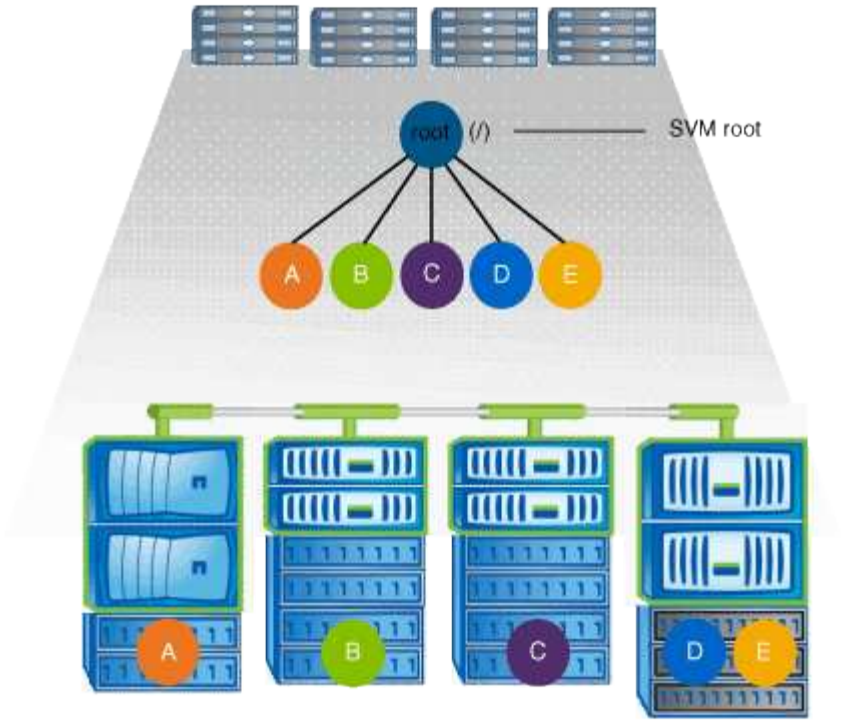



例如、具有上述命名空間架構的典型Volume交會組態可能類似下列組態、其中有三個插入點可插入SVM的根Volume。兩個插入點是名為「dATA」和「專案」的目錄。其中一個插入點是名為「稽核」的輔助磁碟區：

```
                      Junction                          Junction
Vserver Volume        Active   Junction Path            Path Source
------- ------------- -------- --------------------     -----------
vs1     audit         true     /audit                   RW_volume
vs1     audit_logs1   true     /audit/logs1             RW_volume
vs1     audit_logs2   true     /audit/logs2             RW_volume
vs1     audit_logs3   true     /audit/logs3             RW_volume
vs1     eng           true     /data/eng                RW_volume
vs1     mktg1         true     /data/mktg1              RW_volume
vs1     mktg2         true     /data/mktg2              RW_volume
vs1     project1      true     /projects/project1       RW_volume
vs1     project2      true     /projects/project2       RW_volume
vs1     vs1_root      -        /                        -
```

## 具有多個獨立磁碟區的命名空間

在具有獨立磁碟區的架構中、每個磁碟區都有一個插入點、指向SVM命名空間的根目錄、但是該磁碟區並未與另一個磁碟區連結。每個磁碟區都有一個獨特的路徑、可以直接連接到根目錄下方、也可以連接到根目錄下方的目錄下。




例如、具有上述命名空間架構的典型Volume交會組態可能類似下列組態、其中有五個插入點指向SVM的根Volume、每個插入點代表一個Volume的路徑。

```
                      Junction                          Junction
Vserver Volume        Active   Junction Path            Path Source
------- ------------ -------- --------------------     -----------
vs1     eng           true     /eng                     RW_volume
vs1     mktg          true     /vol/mktg                RW_volume
vs1     project1      true     /project1                RW_volume
vs1     project2      true     /project2                RW_volume
vs1     sales         true     /sales                   RW_volume
vs1     vs1_root      -        /                        -
```

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